

data from PCI-based peripherals to and from the memory bus for further processing. This general-purpose bus has been serving well as a standard for over 10 years, hosting such devices as soundcards, video cards and storage controllers. Now PCI is facing a major strain with bandwidth-hungry I/O devices on the horizon. Products used in applications such as video capture/editing and other bandwidth-intensive applications need plenty of room to function. In order to host a number of components such as these, either the entire PCI bus would have to be reworked or a new standard would have to be built from the ground up.

A chipset, or core logic, is responsible for directing the control and flow of data across bus lines. Generally, it is divided into two distinct chips, a north bridge and a south bridge. The term 'bridge' comes from a reference to a device that connects multiple buses together. The north bridge connects to the CPU, the memory bus, the AGP graphics bus and the south bridge. The south bridge connects to the PCI, USB, ATA, ISA, etc.

With current technology, the reservoir of available bandwidth for these multiple buses is quickly running out, leaving the industry no other choice but to innovate new ways to replenish the bandwidth pool.

The answers

One answer to bus bandwidth constraints is HyperTransport, an AMD-led architecture developed with the help of industry partners such as Cisco, nVidia, Sun Microsystems and 168 other industry influences. HyperTransport is a bi-directional (send and receive), high performance packets-based protocol for interconnecting various components in a PC. Using packets simply means sending data in chunks (or packets) and rearranging them to its original form later. It was designed for applications such as networking, telecommunications and other uses where scalability, high speeds and low latencies are required while at the same time, remaining compatible with legacy PC buses such as PCI, ISA, USB, etc.

There are several immediate advantages of HyperTransport. While developing specifications for the protocol, design engineers kept a framework of goals in mind in order to improve system performance. Higher bandwidth bus links and reducing low latency responses across bus points are two direct ways of achieving this.

Being 32-bits wide and running at a speed of 33 MHz, the PCI is restricted to a maximum bandwidth of 133 MBps. HyperTransport offers substantially larger bandwidth up to a maximum of 12.8 GBps, a near 100-fold increase over existing technology.

Current PCI technology transfers data across

the bus at 133 MBps. PCI-X, an evolutionary modification of the original PCI bus, runs along at 1 GBps. HyperTransport ups the ante to a significantly higher 12.8 GBps.

Off the bat, HyperTransport uses a flexible bus architecture providing data links in various widths. Depending on the application, bit widths can either be 2, 4, 8, 16 or 32 bits. Each bus line is capable of transferring data at a rate of 400 MHz in each direction. Since HyperTransport provides a dedicated line for both sending and receiving data, effective throughput is in the area of 800 MHz with a maximum throughput that is over 96 times the bandwidth of today's PCI bus. In situations requiring different upstream and downstream speeds, HyperTransport is fully capable of asymmetric operation.

Optimising bus topology and reducing the number of buses in a system not only streamlines data transfer processes, but also contributes to lower voltage requirements and, therefore, lower power consumption.

HyperTransport is not designed to eliminate technologies such as USB and FireWire. Instead, it was designed to complement them by acting as an infrastructural backbone between the devices themselves and the rest of the system. The fact remains that HyperTransport was designed to interconnect internal system components and not directly connect to external technologies. The point of using HyperTransport is to provide the bandwidth that FireWire and USB devices will need now and in the future.

Unlike PCI, which is a shared bus and allows multiple devices to be placed on the one bus, HyperTransport only allows one device to be placed on a bus. However, these busses can be chained together to create a multi-device bus.

Latencies

Of course, the further devices are chained, the higher latency there will be between a device at the end of the chain and the processor. It is important to design the system so that there are at most around three to five elements in the chain. The more a component is affected by latency, the higher up the chain of buses it has to go, otherwise, a bottleneck would be introduced.

HyperTransport's theoretical maximum of 12.8 GBps is 96 times as fast as a 32-bit, 33 MHz PCI bus, and 24 times faster than an expensive 64-bit, 66 MHz PCI bus. However, the 12.8 GBps link is likely to be used only in processor-north bridge and north bridge-north bridge configurations, and each pin-

Computing at Light Speed

US-based startup, Primarion, is currently in the process of developing optical fibers which will serve as the next medium for bus data transfers. Once processors start touching speeds in excess of 10 GHz, industry analysts say that copper would have reached the limit of its bandwidth carrying capacity. Optical links pose a number of advantages over today's current carrying materials and promise that the dream of video-on-demand and peer-to-peer computing will become a reality.

Advances in processor technology have surpassed the seemingly outdated limits of Moore's Law. As transistor counts continue to skyrocket so does the amount of voltage required to power them. And high amounts of power can do funny things like causing interference in bus signals and lead to physical damage of components.

Optical connections use pulses of laser beams to transmit data as opposed to the electromagnetic principle used in today's technology. Using optics would not only eliminate EMI (Electromagnetic Interference), but would also dramatically bring down power requirements.

Likeness in Disparity		
	HyperTransport	3GIO
Bus type	Point-to-point	Point-to-point
Protocol	Packet based	Packet based
Maximum bandwidth per link	1.6Gbits/sec	2.5Gbits/sec
Multi-processor support	Yes	Yes
Outside-the-box configuration	No	Yes